

Rajabharana Jewellery Management System

Materials & Inventory Module — Sequence Diagrams

Raw Materials · Bill of Materials · Stock Movements · Order Stock Deduction

Status: Implemented ☒ | **Stack:** Laravel 12 · Blade · MySQL · InventoryService

Document: MATERIALS_SEQUENCE_DIAGRAM.pdf

Figure 1 — Add Raw Material (UC-M02)

Inventory Manager creates a new workshop material. Code is auto-generated as RM-YYYYMMDD-XXXX.

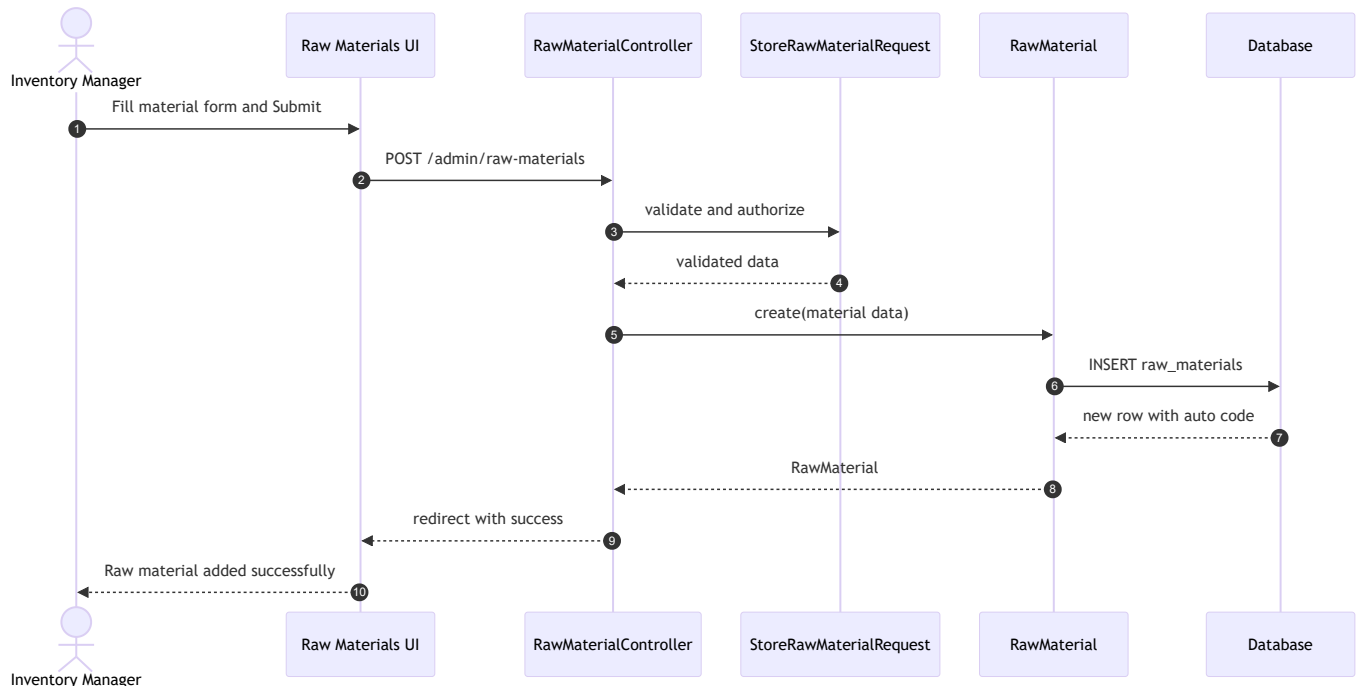


Figure 2 — Adjust Raw Material Stock (UC-M05)

Manual stock increase or decrease. Creates a stock_movements audit record with reason material_received or workshop_usage.

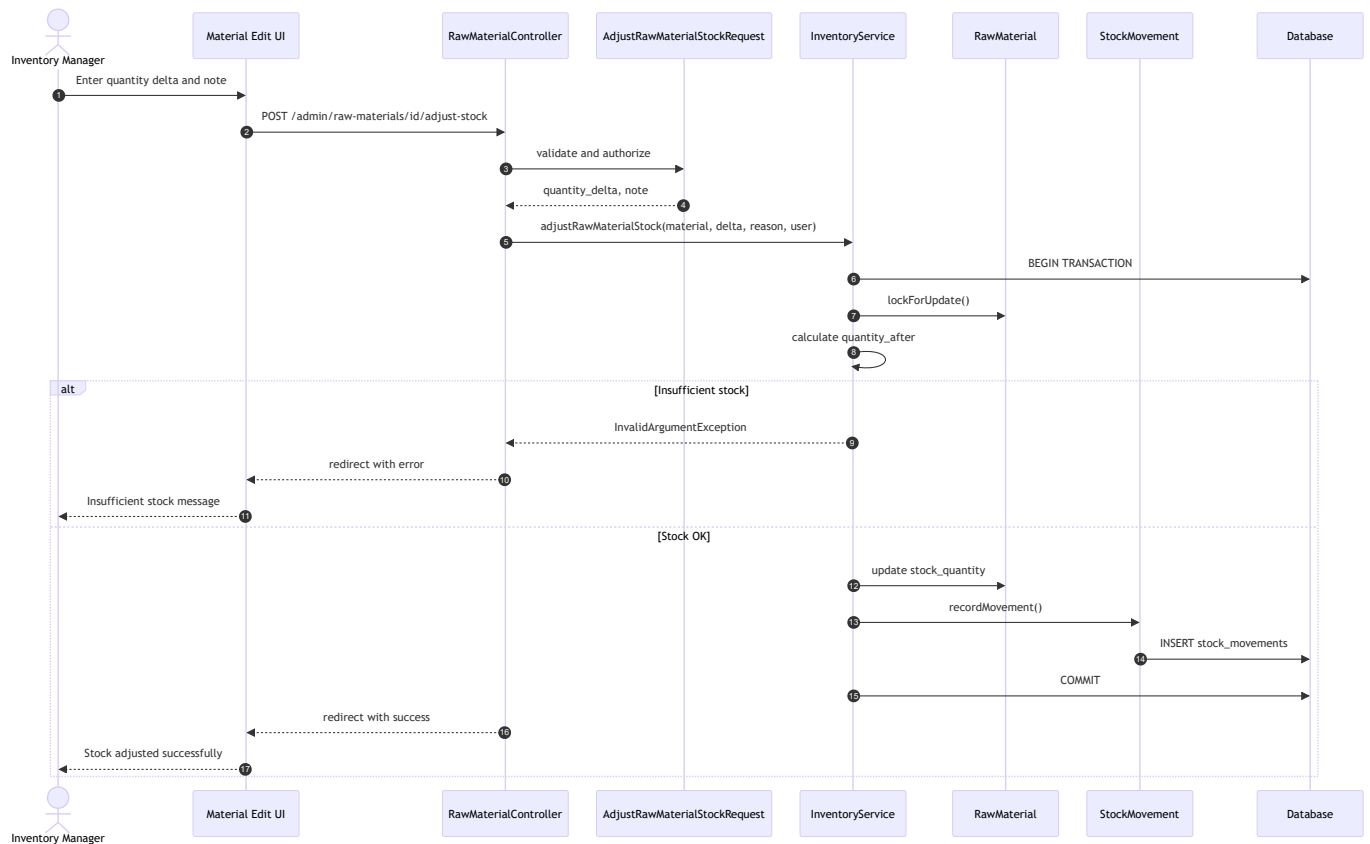


Figure 3 — Link Materials to Catalog Item (UC-M06 + UC-M07)

Manager sets catalog stock quantity and Bill of Materials (quantity_required per material per 1 catalog unit).

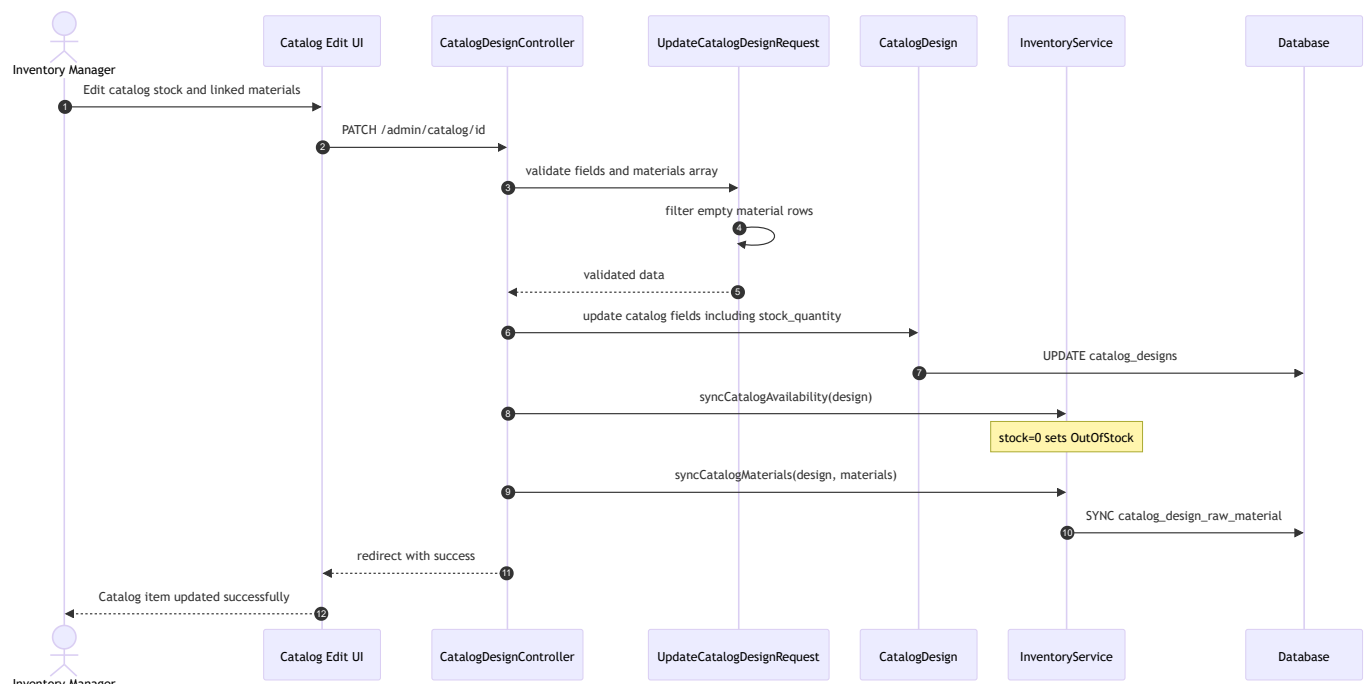


Figure 4 — Customer Places Catalog Order (UC-M11)

Stock is validated at order time but not deducted. Order is saved as Pending.

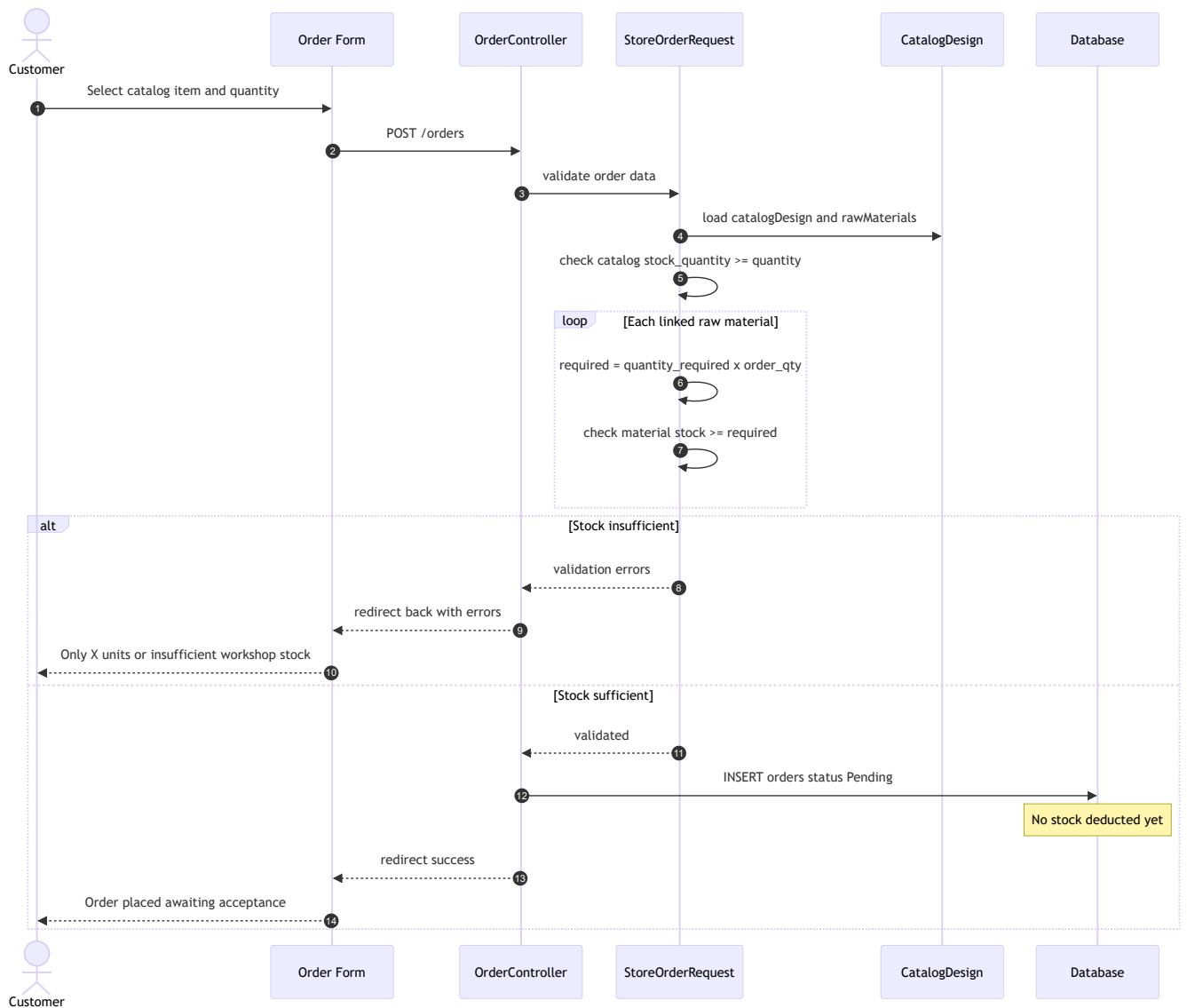


Figure 5 — Accept Catalog Order — Auto Deduct Stock (UC-M09)

Admin accepts order. InventoryService deducts catalog stock and all linked raw materials. Audit logged in stock_movements.

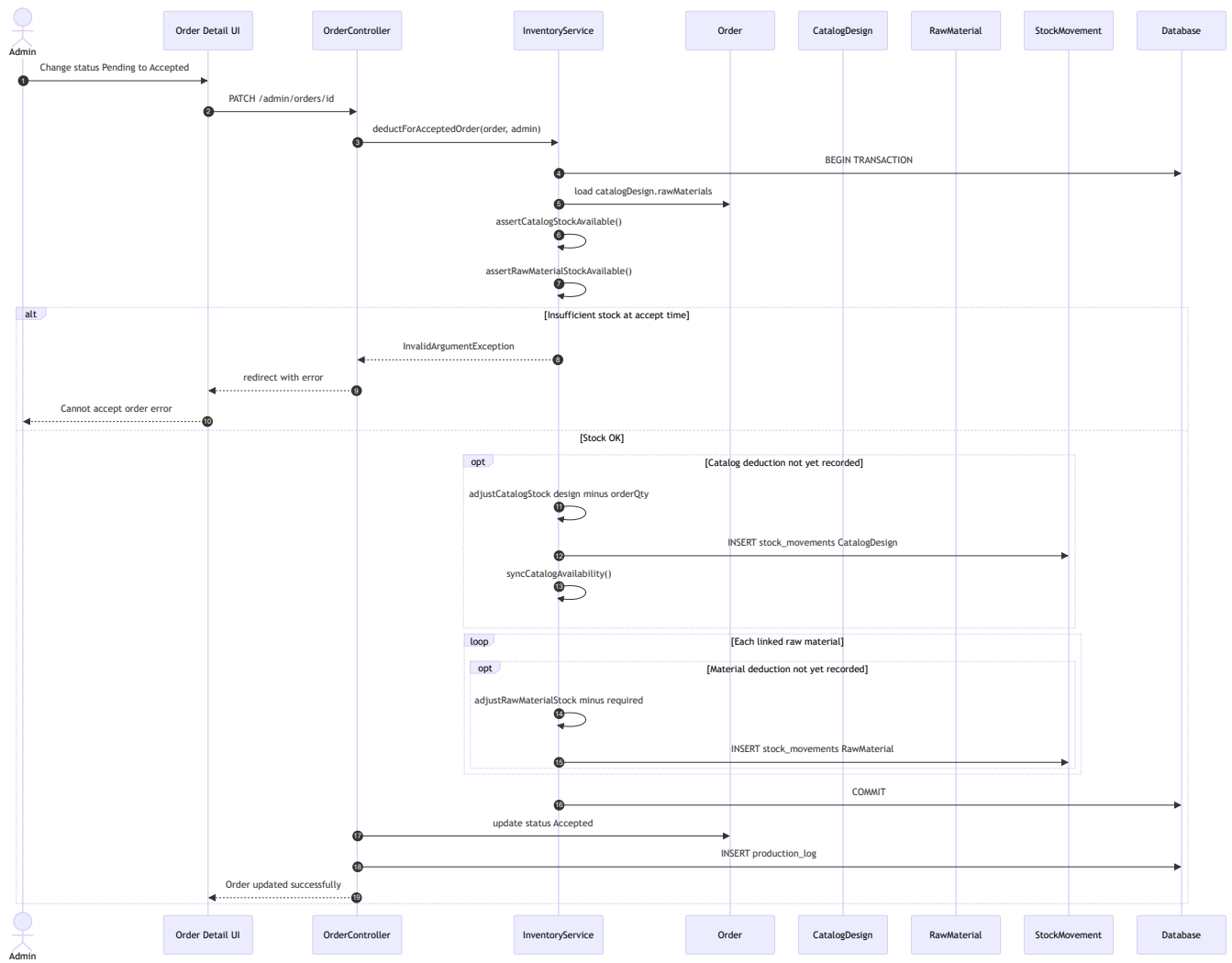


Figure 6 — Reject Accepted Order — Restore Stock (UC-M10)

If an accepted order is rejected, catalog stock and linked materials are restored with audit records.

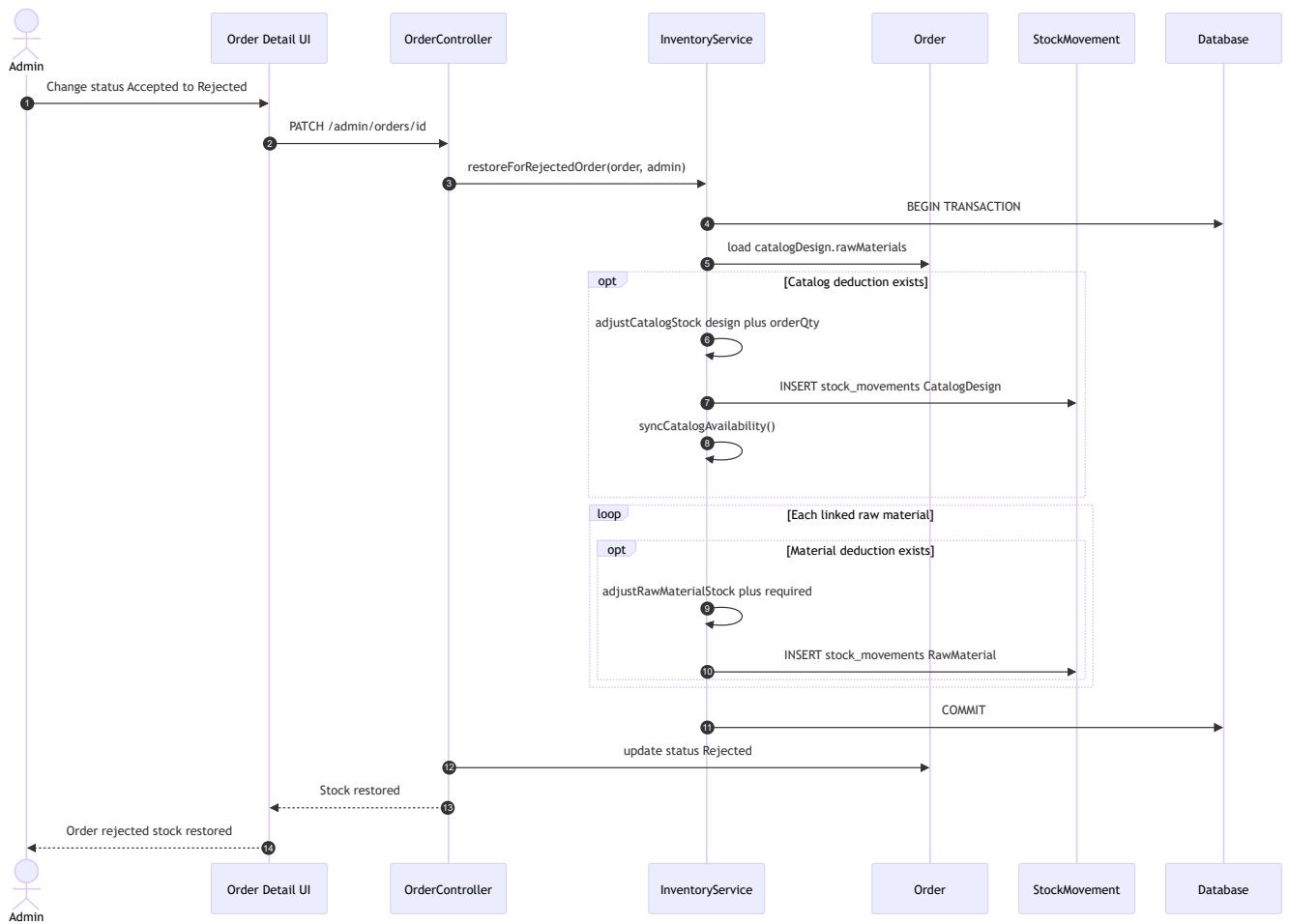


Figure 7 — Generate Inventory Report (UC-M12)

Inventory Manager views category-wise catalog stock KPIs and raw materials summary including low-stock count.

